

Does a Currency Hedge Neutralize Daily Yen Exposure?

货币对冲能否抵消日元的日度风险敞口？

A Multi-Factor Study of HEWJ versus EWJ Before and After the COVID-19 Break

疫情前后 HEWJ 与 EWJ 的多因子比较研究

Course / 课程	STA302 Final Project - Part 1 / STA302 期末项目第一部分
Prepared / 日期	September 20, 2026 / 2026 年 9 月 20 日
Group / 小组	Haiwen Yi; [add all other members / 补充其他成员]
Data / 数据	2,753 daily observations 2014-02-06 to 2026-07-31

English proposal followed by a Chinese study copy / 英文正式提案及中文组内版本

Official English proposal

Contribution statement

Member	Proposed contribution
Haiwen Yi	Research design, data construction, R analysis, diagnostics, and first draft
[Member 2]	Literature verification, interpretation, and written revision
[Member 3]	Reproducibility check, poster development, and presentation recording
[Member 4, if applicable]	Final-model sensitivity analysis and presentation review

Names and responsibilities must match the separately submitted Group Teamwork Agreement.

Introduction (388 words)

International equity investors receive returns from the local equity market and from the exchange rate used to translate foreign assets into their home currency. Currency-hedged exchange-traded funds attempt to remove the second component, but hedging may be incomplete because forward contracts are rolled periodically, trading calendars differ, expenses and implementation costs exist, and portfolios are not rebalanced continuously. This project asks: How strongly does daily yen appreciation explain the return difference between the currency-hedged iShares MSCI Japan ETF (HEWJ) and its unhedged counterpart (EWJ), after controlling for equity, factor, volatility, and interest-rate conditions, and did that relationship change after the World Health Organization characterized COVID-19 as a pandemic?

The response is the daily log-return spread, $Y_t = 100[\Delta \log(\text{HEWJ}_t) - \Delta \log(\text{EWJ}_t)]$. The focal predictor is $\text{JPYapp}_t = -100\Delta \log(\text{DEXJPUS}_t)$, so a positive value denotes yen appreciation against the U.S. dollar. A difference in means or a simple correlation cannot simultaneously adjust for market conditions, estimate conditional period-specific slopes, and test whether the slope changed. Multiple linear regression can do all three through coefficients, confidence intervals, and an interaction test. Under a complete contemporaneous hedge, the yen slope should be near -1. We estimate pre- and post-pandemic slopes with a categorical period indicator and its interaction with yen appreciation. March 11, 2020, is treated as the transition date and omitted.

Four peer-reviewed studies motivate the specification. In international equity and currency data, Hau and Rey (2006) find exchange rates, equity returns, and portfolio flows move jointly, motivating equity-market controls. Using international asset-return data, Campbell, Serfaty-de Medeiros, and Viceira (2010) show that optimal currency hedges vary across investors and assets, so a mechanical one-for-one hedge need not hold. Across diversified U.S. stock portfolios, Fama and French (2015) show that size, value, profitability, and investment factors explain average-return variation, supporting factor controls. For U.S.-listed currency-hedged ETFs, Shank and Vianna (2016) find dynamic links between exchange rates and investor trading. Our contribution is a reproducible daily matched-pair analysis of one Japan ETF pair with an explicit structural-break interaction and time-ordered data.

The results would benefit U.S.-dollar investors, portfolio managers, and risk teams deciding whether a “hedged” Japan allocation actually offsets daily yen exposure. The pandemic indicator is descriptive, not causal, because many policies and market conditions changed simultaneously.

Data description (276 words)

Variable	Type	Mean	SD	Range	Important feature
Response spread	Numerical	0.021	0.627	-4.310 to 3.783	Extreme days; heavy tails
Yen appreciation	Numerical	-0.014	0.585	-2.850 to 4.322	Wide daily extremes
Nikkei return	Numerical	0.045	1.366	-13.234 to 9.737	Large crash/rebound days
Japan SMB	Numerical	-0.002	0.498	-2.340 to 3.120	Moderate spread
Japan HML	Numerical	0.018	0.723	-3.590 to 4.570	Relatively wide factor range
Japan RMW	Numerical	-0.003	0.414	-1.840 to 2.220	Moderate spread
Japan CMA	Numerical	0.008	0.388	-2.370 to 2.290	Moderate spread
Japan MOM	Numerical	-0.012	0.810	-7.150 to 8.550	Wide factor extremes
VIX log change	Numerical	-0.053	8.108	-44.245 to 76.825	Largest dispersion and right extreme
Lagged rate difference	Numerical	1.853	1.828	-0.010 to 5.394	Slow-moving monthly series
Post period	Categorical	Pre: 1,348	Post: 1,405	2 levels	WHO-defined break

Table 1. Numerical summary of the response and every model predictor. Units are percentage points except the categorical period indicator.

The analysis contains 2,753 complete daily observations from February 6, 2014, through July 31, 2026. Yahoo Finance distributes histories assembled from market quotations and corporate actions for investment analysis; we use adjusted closes for HEWJ and EWJ. FRED redistributes official and market series for economic research: DEXJPUS is the Federal Reserve's noon New York yen-per-dollar quote; the Nikkei 225, VIX, and short-term rates represent equity, expected volatility, and monetary conditions. Kenneth French's Data Library constructs research portfolios from security returns using published factor definitions; we use its Japan five-factor and momentum series.

The response, the daily HEWJ-minus-EWJ log-return spread, has mean 0.0206 percentage points, standard deviation 0.6271, and range -4.3097 to 3.7826. It is continuous and can take either sign, making a Gaussian linear conditional mean interpretable. Daily observations are temporally ordered rather than strictly independent, so independence is explicitly assessed in the residual analysis and no causal interpretation is made.

Table 1 summarizes the response and every predictor. Extreme daily movements appear in the response, Nikkei return, momentum, and VIX change; the rate differential is slow-moving because monthly observations are carried forward with a one-month lag. Post has 1,348 pre-period and 1,405 post-period observations. Complete cases are retained after joining by trading date.

The model includes yen appreciation, Post, Nikkei return, SMB, HML, RMW, CMA, MOM, log VIX change, and the lagged U.S.-Japan rate differential. The $\text{JPYapp} \times \text{Post}$ interaction directly answers whether the conditional yen slope differs across the pandemic break. The split is substantively pre-specified from the WHO announcement rather than chosen by searching for the smallest p-value. Data websites and complete citations appear below.

Ethics discussion (189 words)

The dataset is trustworthy because its provenance, ownership, and transformations are documented. Yahoo Finance closes, FRED series, and Kenneth French factors are public sources, but authority does not eliminate error: providers can revise observations, calendars may differ, and adjusted prices depend on provider methods. We preserve snapshots and SHA-256 checksums, document transformations, and avoid overstating precision. This addresses the module's trustworthiness criteria of provenance, transparency, accuracy, and reproducibility.

Collection and use are ethically low risk. The sources contain aggregate market prices and factor portfolios, not personal information, private holdings, individual trading records, or human-subject data. Consent, confidentiality, and re-identification risks are minimal. We respect ownership by citing each provider, keeping the repository private while terms are reviewed, and not claiming third-party observations.

The principal ethical risk is interpretation. A descriptive association could be presented incorrectly as investment advice or evidence that COVID-19 caused a change. We will state that results apply to one ETF pair, acknowledge survivorship and product-selection limitations, and avoid causal language because changes were uncontrolled. Historical predictive performance is not a promise of future returns. These limits make the use proportionate to the data and question.

Preliminary results (351 words)

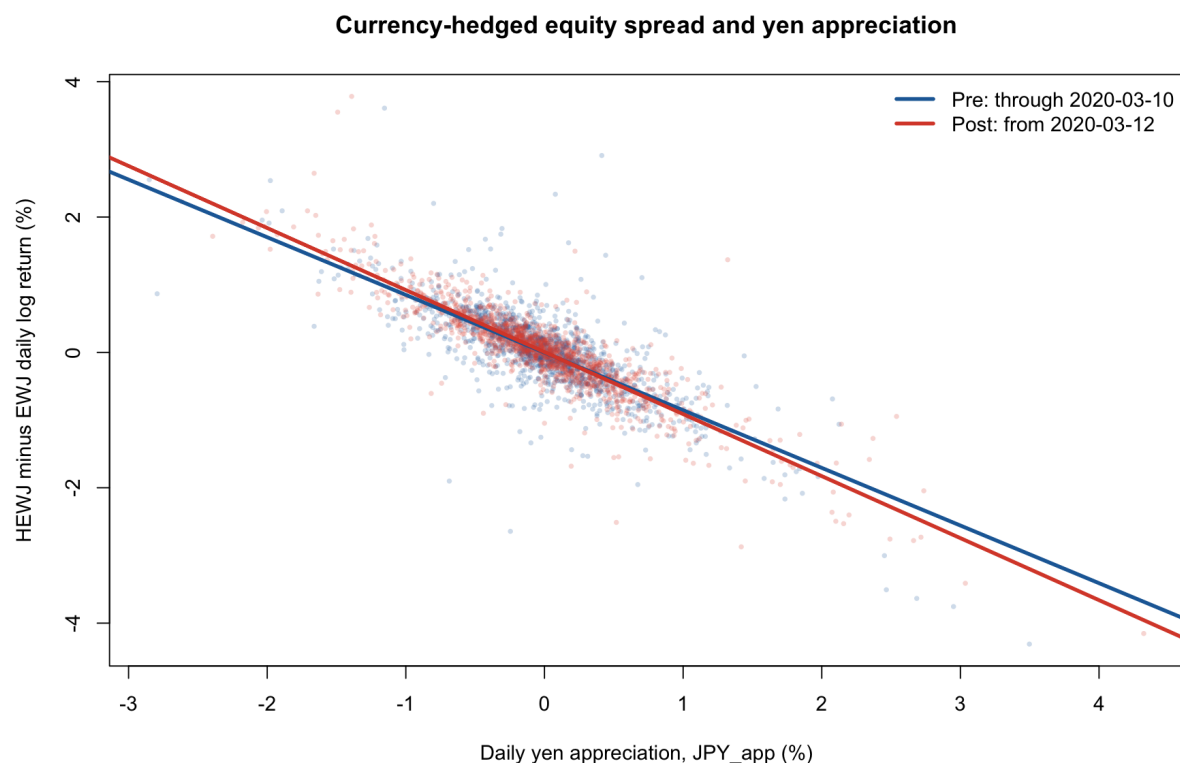


Figure 1. Daily HEWJ-minus-EWJ return spread versus yen appreciation, with controlled period-specific OLS slopes.

Term	Estimate	Classical SE	95% CI	p
Intercept	-0.0018	0.0100	[-0.0213, 0.0178]	0.859
Yen appreciation	-0.8520	0.0178	[-0.8868, -0.8171]	<0.001
Post indicator	0.0085	0.0147	[-0.0203, 0.0373]	0.562
Nikkei return	-0.0042	0.0059	[-0.0158, 0.0073]	0.472
SMB	0.0208	0.0149	[-0.0085, 0.0501]	0.164
HML	-0.0216	0.0167	[-0.0544, 0.0112]	0.197
RMW	-0.0084	0.0267	[-0.0608, 0.0440]	0.754
CMA	0.0378	0.0239	[-0.0090, 0.0847]	0.114
MOM	0.0021	0.0086	[-0.0148, 0.0190]	0.805
VIX log change	-0.0069	0.0008	[-0.0085, -0.0053]	<0.001
Lagged rate difference	0.0028	0.0040	[-0.0051, 0.0106]	0.493
Yen x Post	-0.0648	0.0227	[-0.1093, -0.0204]	0.004

Table 2. Complete preliminary OLS coefficient table with classical standard errors and 95% confidence intervals.

The full ordinary least-squares model is

$$Y(t) = b_0 + b_1 JPYapp(t) + b_2 Post(t) + b_3 [JPYapp(t) \times Post(t)] + g'Z(t) + e(t),$$

where Z_t contains the eight market and factor controls. Table 2 reports every coefficient with a classical standard error and 95% confidence interval. The model explains 71.21% of daily spread variation (adjusted $R^2=71.10\%$). The pre-period yen slope is -0.852 (SE 0.0178; 95% CI [-0.887, -0.817]). Thus, conditional on the controls, a one-percentage-point yen appreciation is associated with a 0.852-percentage-point lower HEWJ-minus-EWJ return. The interaction is -0.0648 (SE 0.0227; $p = 0.0043$), giving an implied post-period slope of -0.917. The period level shift is not significant. VIX log change is negative; most other controls are imprecisely estimated.

The strong negative currency slope is consistent with Hau and Rey's joint currency-equity mechanism and with the hedge interpretation. Its departure from exactly -1 agrees with Campbell et al.'s result that effective hedges need not be mechanical one-for-one positions. The mostly small factor coefficients suggest that pairing HEWJ with EWJ removes much common equity exposure, while retaining Fama-French controls prevents that conclusion from being assumed. The changed slope is directionally compatible with Shank and Vianna's evidence of time-varying currency-hedged ETF dynamics, although our pandemic comparison is not causal.

Figure 2 provides the complete preliminary diagnostic grid. Residuals versus fitted values show no dominant smooth curve, but the RESET test rejects exact functional form ($p = 0.001$), so linearity remains questionable. Residual spread changes across fitted values and the Breusch-Pagan test rejects constant variance ($p = 0.0048$). The Q-Q plot has heavy tails and Jarque-Bera rejects normality ($p < 0.001$). Residuals over time and the ACF show dependence; the five-lag Breusch-Godfrey test rejects independence ($p < 0.001$), and Durbin-Watson is 2.85. The maximum VIF is 3.55, so severe multicollinearity is not evident. There are 137 cases above the Cook's-distance screening threshold $4/n$, with maximum 0.136, indicating influential observations for later investigation.

These are preliminary OLS results. In accordance with the Part 1 instruction, we diagnose but do not correct violations here; robust inference and sensitivity analysis are deferred to the final project.

Residual diagnostics

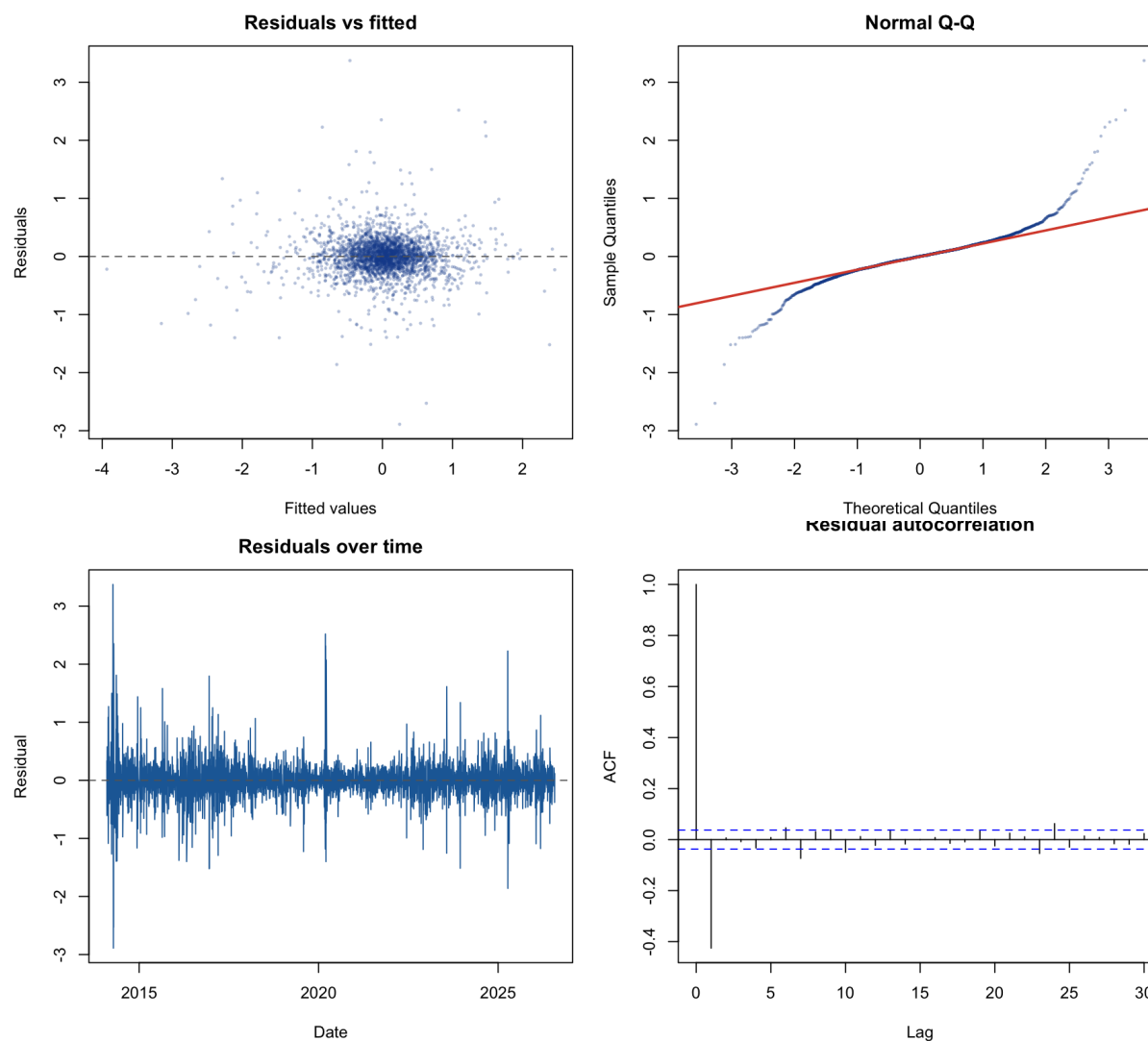


Figure 2. Residual-versus-fitted, Q-Q, time-order, and autocorrelation plots for the uncorrected preliminary OLS model.

Plan for the remaining analysis (275 words)

The focal terms - yen appreciation, Post, and their interaction - will remain because they define the research question. Other controls will first be retained on substantive grounds. We will compare theory-preserving nested models using partial F-tests, adjusted R^2 , residual behavior, and focal-coefficient stability. We will not drop a control solely because its p-value exceeds 0.05, and hierarchy will be respected.

After every material specification change, we will recheck residual-versus-fitted, Q-Q, time-order, and ACF plots, Breusch-Pagan and Breusch-Godfrey tests, Cook's distance, and VIF. Because the model shows heteroskedasticity, dependence, heavy tails, and functional-form evidence, we will examine a pre-specified nonlinear yen term and a Yeo-Johnson response transformation, which is valid for signed returns; a direct logarithm is invalid. Transformations will be compared for assumption improvement and interpretability, not selected to manufacture significance.

Influential dates will be documented before any sensitivity run. The primary analysis will retain all observations; secondary analyses will compare declared 0.5/99.5-percentile winsorization and the exclusion of individually justified data errors, if any. Once a final functional form is chosen, classical intervals will be supplemented by Newey-West intervals with 1-, 5-, and 10-day lags. We will also vary the pandemic cut within a small declared window rather than search all dates.

Prediction remains secondary. A chronological 80/20 split will train on earlier dates and evaluate later dates against a historical-mean benchmark using RMSE and MAE. No random cross-validation will be used. The frozen raw files, cleaned CSV, code, and outputs will be versioned together.

The schedule in Table 3 assigns analysis, poster, and recording milestones. Member placeholders will be replaced with names from the signed Teamwork Agreement before submission.

Dates	Milestone	Owner(s)	Checkable output
Nov 9-15	Audit cleaned data and diagnostics	Haiwen + Member 3	Frozen data; issue log
Nov 16-22	Revise model and run sensitivities	Haiwen + Member 2	Final model tables/figures
Nov 23-27	Create and review poster	Member 2 + Member 3	Complete poster draft
Nov 28-30	Record, caption, and review	All members	Final recording
Dec 1-2	Final file and link audit	Haiwen + all members	Submission-ready package

Table 3. Proposed team schedule. Member placeholders must be replaced with the signed roster.

References

- Campbell, J. Y., Serfaty-de Medeiros, K., & Viceira, L. M. (2010). Global currency hedging. *The Journal of Finance*, 65(1), 87-121. <https://doi.org/10.1111/j.1540-6261.2009.01524.x>
- Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1-22. <https://doi.org/10.1016/j.jfineco.2014.10.010>
- Hau, H., & Rey, H. (2006). Exchange rates, equity prices, and capital flows. *The Review of Financial Studies*, 19(1), 273-317. <https://doi.org/10.1093/rfs/hhj008>
- Shank, C. A., & Vianna, A. C. (2016). Are US-dollar-hedged-ETF investors aggressive on exchange rates? A panel VAR approach. *Research in International Business and Finance*, 38, 430-438. <https://doi.org/10.1016/j.ribaf.2016.05.002>

Data and product documentation

- Federal Reserve Bank of St. Louis. (n.d.). DEXJPUS; NIKKEI225; VIXCLS; IRSTCI01USM156N; IRSTCI01JPM156N [Data sets]. FRED. <https://fred.stlouisfed.org/>
- iShares. (n.d.). HEWJ and EWJ fund pages and currency-hedged product documentation. <https://www.ishares.com/>
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- World Health Organization. (2020, March 11). WHO Director-General's opening remarks at the media briefing on COVID-19. <https://www.who.int/>
- Yahoo Finance. (n.d.). HEWJ and EWJ historical data [Data sets]. <https://finance.yahoo.com/>

Submission items requiring group confirmation

1. Replace bracketed member names and role assignments with the actual group roster.
2. Complete and sign the course's Group Teamwork Agreement PDF using the same names and responsibilities.
3. Upload the prepared original and cleaned CSV folders to UofT OneDrive and add a Quercus submission comment with permission set to "anyone in UofT with the link can access."

第二部分 - 中文提案（组内讨论与理解版本）

正式课程提交请使用独立英文 PDF；本部分与英文版采用相同研究设计、数值与限制。

成员贡献说明

成员	计划贡献
Haiwen Yi	研究设计、数据构建、R 分析、诊断检验和初稿
[成员 2]	文献核验、结果解释和文字修订
[成员 3]	可复现性检查、海报制作和演示录制
[成员 4, 如适用]	最终模型敏感性分析与演示审核

姓名与职责必须和单独提交的小组协议一致。

研究背景与问题

国际股票投资者的收益同时受到本地股票市场和汇率影响。货币对冲 ETF 试图消除汇率部分，但由于远期合约展期、交易日差异、费用和非连续再平衡，对冲可能不完整。本项目研究：在控制股票市场、资产定价因子、波动率和利率条件后，日元升值在多大程度上解释货币对冲型日本股票 ETF（HEWJ）与非对冲型 ETF（EWJ）的日度收益差？这种关系在世界卫生组织宣布 COVID-19 为全球大流行后是否改变？

响应变量是日度对数收益差 $Y_t = 100[\Delta \log(\text{HEWJ}_t) - \Delta \log(\text{EWJ}_t)]$ ，核心预测变量为 $\text{JPYapp}_t = -100\Delta \log(\text{DEXJPUS}_t)$ 。简单均值比较或相关系数无法同时控制市场条件、估计分时期条件斜率并检验斜率是否改变；多元线性回归可以通过系数、置信区间和交互项检验回答这些问题。完整同期对冲下，日元斜率应接近 -1。模型以分类变量 Post 及其与日元升值的交互项估计疫情前后斜率，并剔除转换日 2020 年 3 月 11 日。

Hau 与 Rey（2006）说明汇率、股票收益与投资组合流动共同变化；Campbell 等（2010）发现最优货币对冲随投资者和资产而异；Fama 与 French（2015）为规模、价值、盈利和投资因子控制提供依据；Shank 与 Vianna（2016）发现美国上市货币对冲 ETF 的交易与汇率存在动态联系。本研究用一个日本对冲/非对冲 ETF 配对进行可复现日度分析。结果可帮助美元投资者、组合经理和风险团队判断“已对冲”配置是否真正抵消日元风险。疫情指标只描述结构变化，不代表因果效应。

数据说明

变量	类型	均值	标准差	范围	主要特征
响应变量收益差	数值	0.021	0.627	-4.310 to 3.783	存在极端日与厚尾
日元升值	数值	-0.014	0.585	-2.850 to 4.322	日度极端值较宽
日经收益	数值	0.045	1.366	-13.234 to 9.737	存在大跌与反弹日
日本 SMB	数值	-0.002	0.498	-2.340 to 3.120	离散程度中等
日本 HML	数值	0.018	0.723	-3.590 to 4.570	因子范围较宽
日本 RMW	数值	-0.003	0.414	-1.840 to 2.220	离散程度中等
日本 CMA	数值	0.008	0.388	-2.370 to 2.290	离散程度中等
日本 MOM	数值	-0.012	0.810	-7.150 to 8.550	存在较宽极端值
VIX 对数变化	数值	-0.053	8.108	-44.245 to 76.825	离散最大且右侧极端
滞后利差	数值	1.853	1.828	-0.010 to 5.394	按月更新、变化缓慢
疫情后时期	分类	疫情前：1,348	疫情后：1,405	2 个水平	WHO 事件定义断点

表 1：响应变量及全部入模预测变量的数值汇总。

分析包含 2014 年 2 月 6 日至 2026 年 7 月 31 日的 2,753 个完整日度观测。Yahoo Finance 汇集市场报价与公司行动记录并发布证券历史数据，本研究使用 HEWJ 和 EWJ 复权收盘价。FRED 为经济研究和公共监测重新发布官方或市场序列，包括纽约中午日元兑美元报价、日经 225、VIX 和短期利率。Kenneth French Data Library 按公开因子定义从证券收益构造研究组合，本研究使用日本五因子和动量因子。

响应变量均值为 0.0206 个百分点、标准差 0.6271，范围为 -4.3097 至 3.7826。它是可正可负连续变量，因此条件均值线性模型具有可解释性；但日度观测并非严格独立，所以残差分析会正式检验时间依赖。表 1 汇总响应变量和全部预测变量。收益差、日经收益、动量和 VIX 变化包含极端日；利差按月更新并滞后一个月。Post 含疫情前 1,348 个和疫情后 1,405 个观测。

模型包含日元升值、Post、日经收益、SMB、HML、RMW、CMA、MOM、VIX 对数变化及滞后美日利差。JPYapp × Post 直接检验疫情断点前后条件日元斜率是否不同；断点来自 WHO 事件，而不是事后搜索最小 p 值。

伦理声明

数据具有可追溯的来源、所有权和变换记录，因此基本可信。Yahoo Finance、FRED 与 Kenneth French 是公开来源，但权威来源仍可能修订观测，交易日历也可能不一致，复权价格还依赖提供方方法。项目保存冻结快照与 SHA-256 校验值、记录全部变换并避免夸大精度，对应伦理模块中的来源可追溯、透明度、准确性和可复现性标准。

数据只包含聚合市场价格和因子组合，不包含个人信息、私人持仓、个人交易或人体研究记录，因此同意、保密和再识别风险很低。项目会引用每个提供方，在条款核验完成前保持仓库私有，并不宣称拥有第三方观测。

主要伦理风险来自解释。描述性关联不应被写成投资建议或 COVID-19 的因果效应。报告会说明结论仅适用于一个 ETF 配对，承认存续偏差和产品选择限制，并避免因果措辞。历史预测表现也不代表未来收益承诺。

初步结果

变量	估计值	经典标准误	95% 区间	p
截距	-0.0018	0.0100	[-0.0213, 0.0178]	0.859
日元升值	-0.8520	0.0178	[-0.8868, -0.8171]	<0.001
疫情后指标	0.0085	0.0147	[-0.0203, 0.0373]	0.562
日经收益	-0.0042	0.0059	[-0.0158, 0.0073]	0.472
SMB	0.0208	0.0149	[-0.0085, 0.0501]	0.164
HML	-0.0216	0.0167	[-0.0544, 0.0112]	0.197
RMW	-0.0084	0.0267	[-0.0608, 0.0440]	0.754
CMA	0.0378	0.0239	[-0.0090, 0.0847]	0.114
MOM	0.0021	0.0086	[-0.0148, 0.0190]	0.805
VIX 对数变化	-0.0069	0.0008	[-0.0085, -0.0053]	<0.001
滞后利差	0.0028	0.0040	[-0.0051, 0.0106]	0.493
日元 x 疫情后	-0.0648	0.0227	[-0.1093, -0.0204]	0.004

表 2：完整初步 OLS 系数、经典标准误与 95% 置信区间。

完整普通最小二乘模型为：

$$Y(t) = b_0 + b_1 \text{JPYapp}(t) + b_2 \text{Post}(t) + b_3 [\text{JPYapp}(t) \times \text{Post}(t)] + g'Z(t) + e(t).$$

表 2 给出全部系数、经典标准误和 95% 置信区间。模型解释 71.21% 的日度收益差变异（调整后 $R^2=71.10\%$ ）。疫情前日元斜率为 -0.852（SE 0.0178；95% CI [-0.887, -0.817]）；交互项为 -0.0648（SE 0.0227； $p=0.0043$ ），得到疫情后隐含斜率 -0.917。时期水平变化不显著，VIX 对数变化系数为负，其他多数控制项估计不精确。

强负斜率符合 Hau 与 Rey 的汇率-股票联动机制；斜率并非严格 -1 与 Campbell 等人的非机械式最优对冲结果一致。多数因子系数较小，说明配对差分消除了大量共同股票风险。斜率变化方向上与 Shank 和 Vianna 的动态关系结果相容，但不能解释为疫情因果效应。

图 2 显示完整初步诊断。残差-拟合值图没有明显单一曲线，但 RESET 检验拒绝精确函数形式（ $p=0.001$ ）；Breusch-Pagan 拒绝同方差（ $p=0.0048$ ）；Q-Q 图厚尾且 Jarque-Bera 拒绝正态性；时间残差图和 ACF 显示依赖，五阶 Breusch-Godfrey 拒绝独立性（ $p<0.001$ ），Durbin-Watson 为 2.85。最大 VIF 为 3.55。共有 137 个观测超过 Cook 距离筛查阈值 $4/n$ ，最大值 0.136。

本部分只诊断、不修正模型违背；稳健推断和敏感性分析留到最终项目。

后续分析计划

日元升值、Post 和交互项定义研究问题，因此必须保留。其他市场、因子、波动率和利率控制先依据理论保留，再通过部分 F 检验、调整后 R^2 、残差表现和核心系数稳定性比较保持理论含义的嵌套模型。不会仅因单个 p 值大于 0.05 而删除变量，并始终遵守交互项层级原则。

每次实质性修改后都会重新检查残差-拟合值图、Q-Q 图、时间图、ACF、Breusch-Pagan、Breusch-Godfrey、Cook 距离和 VIF。针对异方差、序列相关、厚尾和函数形式证据，将预先比较非线性日元项和可处理负值的 Yeo-Johnson 响应变换；不能直接对可为负响应变量取对数。变换依据假设改善和可解释性选择，而不是为了制造显著性。

主分析保留所有观测；次要分析才使用预先声明的 0.5/99.5 分位缩尾或删除经证实的数据错误。最终函数形式确定后再加入 1、5、10 日 Newey-West 区间，并在小范围事先声明窗口内改变疫情断点。预测只作为次要检查，采用按时间 80/20 划分，与历史均值基准比较 RMSE 和 MAE，不做随机交叉验证。

表 3 安排分析、海报和录制里程碑。提交前需把成员占位符替换成签署协议中的真实姓名。

日期	里程碑	负责人	可检查输出
11 月 9-15 日	复核清理数据与诊断	Haiwen + 成员 3	冻结数据、问题日志
11 月 16-22 日	模型修订与敏感性分析	Haiwen + 成员 2	最终模型表与图
11 月 23-27 日	制作并审核海报	成员 2 + 成员 3	完整海报草稿
11 月 28-30 日	录制、字幕与复核	全组	最终视频
12 月 1-2 日	文件与链接最终检查	Haiwen + 全组	可提交包

表 3：拟定团队进度。提交前必须替换成员占位符。

参考文献

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提交前仍需确认

1. 用真实小组名单替换成员和职责占位符。
2. 完成并签署课程 Group Teamwork Agreement PDF，确保姓名与职责一致。
3. 把准备好的原始和清理 CSV 文件夹上传至 UofT OneDrive，并在 Quercus 评论中加入链接，权限设为 “ anyone in UofT with the link can access ”。